Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	XH-Cat
Selected Domain	1
Selected SDG Track (1–17)	8,4
Project / Solution Name	FutureReady
Name (Leader)	Yip Gui Rong
Email & Phone (Leader)	guirong05@gmail.com 011-61511138
University Affiliation(s)	Multimedia University
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian diploma and foundation students choose a course and a first job without knowing which skills AI is changing. They are among the most exposed: the youth unemployment rate (ages 15 to 24) was 10.2% in June 2026 (OpenDOSM), and entry-level tasks are where AI use is spreading fastest. Training is not keeping up: 56% of Malaysian professionals use AI at work but only 26% have formal training (Hays Asia Salary Guide 2025, quoted in the SDC Domain 1 brief), and more than 600,000 workers need reskilling within 3 to 5 years (MOSTI, quoted in the brief).

Students today get generic career advice or a chatbot that guesses. The official facts exist but are scattered: the national occupation list (eMASCO), job-market statistics (OpenDOSM), and AI-exposure research (ILO, OECD). A student cannot join them into one answer to the question that matters: "I have these skills. Which future roles fit me, and what should I learn first?"

Why it matters: a wrong choice costs a student time and money, and employers and the country lose talent. The students with the least family or industry guidance are hurt most. The problem sits under SDG 4 (relevant skills for decent jobs, target 4.4) and SDG 8 (decent work and productivity through technology, targets 8.2 and 8.6). Our first users are final-year diploma and foundation students; next are working adults at risk, HR teams and TVET career centres.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureReady MY is a working web app that turns a student's skills into ranked roles, a clear view of how AI affects each role, and an ordered learning pathway with real courses.

How it works. (1) The student picks skills or uploads a PDF CV, which is read in the browser. BAAI/bge-m3, an open multilingual embedding model that also reads Bahasa Malaysia, matches CV lines to skills; the student confirms every AI suggestion. (2) Roles, tasks and skills come only from eMASCO (1,766 roles). (3) Current Skill Match is a fixed formula: how well the student's skills, at their stated level, cover the role's required skills. It is explained skill by skill. (4) Beside it we show job outlook (OpenDOSM), observed AI use per role (Anthropic Economic Index, matched to eMASCO tasks by embeddings), and ILO and OECD AI-exposure views, each with source and date. (5) Gaps become an ordered pathway with the exact HRD Corp e-Latih or MyTVET course page, and a what-if shows how the match could change. (6) Every role and skill is tagged to SDGs.

Where AI is used: embeddings for CV, skill and task matching, and optional model tagging of an information-only news feed. A language model never produces a score. The app supports the student's decision and never decides.

Workshop skills applied: embedding and GenAI model use, cloud GPU (RunPod) to compare five embedding models, a file-based data architecture with automatic refresh, and cloud hosting.

Status: the prototype is built and tested (1,011 app tests and 264 data-pipeline tests passing).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology
Frontend	React 19, Vite 8, Tailwind CSS 4 (single-page app, readable at 360 px)
Backend	Node 22 and Express 5 (one app serves the web page and the API)
Data / database	JSON files, rebuilt by scripts; refreshes are atomic and keep the last good file (no database server)
AI	BAAI/bge-m3 multilingual embeddings run locally through Transformers.js (ONNX), or a remote embedding service; used for CV lines, skill search and task matching. No generative model produces any score.
Pipeline	Python 3 (standard library only) to crawl and clean eMASCO; Node scripts to embed, match tasks and build AI-exposure data
Open data and APIs	eMASCO (MOHR), OpenDOSM, OpenAlex, Semantic Scholar, arXiv, Hugging Face (Anthropic Economic Index), ILO and OECD publications, HRD Corp e-Latih and MyTVET course pages
Quality	Vitest (1,011 tests) and Python unittest (264 tests); weekly GitHub Actions data refresh
Hosting	Local Node server plus Cloudflare tunnel for demos; Replit deployment from the main branch

No sponsor GenAI API was used. The GPU credit from the organisers on RunPod was used once, on 22 Oct 2026 (one A40 Pod, secure cloud, USD 0.59/hour), to compare five embedding models on 30 labelled skill pairs; the app does not need RunPod at run time.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: our user is a final-year diploma or foundation student. We used the brief's evidence (Hays Asia, MOSTI) and OpenDOSM youth data to understand the need.

Define: "I have skills today. Which future roles fit me, and what do I learn first?"

Ideate: we rejected "upload a CV and ask a chatbot". Instead: fixed formula, open data, every number with a source and date, and a student who decides.

Prototype: a five-step journey (About you, Your skills, Your goal, Matches, What to study), sample students and made-up CVs only, readable at 360 px, 44 px touch targets, reduced-motion support.

Test: five embedding models compared on 30 labelled skill pairs; CV matching measured on 12 made-up CVs (held-out precision 0.86, recall 0.60, reported openly); a fairness test; an independent QA of the scoring maths; 1,011 automated tests.

Deployment: one Node and Express app (npm run build, npm start), deployable on Replit from the main branch. Every AI step has a non-AI fallback, and data refreshes keep the last good file and show its date. Phase 2 adds a pilot with real students.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4, Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." SDG 8, Target 8.6: "By 2020, substantially reduce the proportion of youth not in employment, education or training." (The app also tags roles to Target 8.2, productivity through technology.)
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	(1) Youth (15 to 24) unemployment rate 10.2%, June 2026 (OpenDOSM, lfs_month_youth). (2) 26% of Malaysian professionals have formal AI training, while 56% use AI at work (Hays Asia Salary Guide 2025, quoted in the SDC Domain 1 brief). (3) Our own starting point: 31 of the 38 skills in the app (82%) have an exact, checked official course page (HRD Corp e-Latih or MyTVET); 7 show only the portal link (data/skills.json, 24 Oct 2026).
Core Measurable KPI	Share of pathway steps that give the student an exact, checked official course page. In the pilot: share of students

Metric Category	Your Response
The specific unit you will track *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	who finish with a ranked role list and a pathway of at least 3 steps, and time taken to get there.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 90% of pathway steps with an exact official course page (from 82%). In a pilot of at least 20 students: 70% finish a pathway in under 10 minutes and 70% say they know which skill to learn first. These are targets, not results; the pilot has not run yet.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, React DOM 19	code	MIT
Vite 8, @vitejs/plugin-react	code	MIT
Tailwind CSS 4, @tailwindcss/vite	code	MIT
Express 5	code	MIT
Transformers.js (@huggingface/transformers 4)	code	Apache-2.0
pdf.js (pdfjs-dist 6.4)	code	Apache-2.0
Vitest 5	code (testing)	MIT
BAAI/bge-m3 (ONNX conversion)	model	MIT
eMASCO occupation list (MOHR, MASCO 2020)	dataset	Public government website; no licence stated; used with

Name	Type (code/dataset/model)	License
		link and snapshot date
OpenDOSM wages_occupation, lfs_month_youth	dataset	CC BY 4.0 (data.gov.my)
ILO Working Paper 140 (2025), Table A1	dataset	CC BY 4.0
OECD AI Papers No. 59 (2026), Table 6.3	dataset	CC BY 4.0
Anthropic Economic Index (release 2026-06-26)	dataset	MIT (dataset card)
OpenAlex counts; Semantic Scholar and arXiv counts	dataset (API)	CC0 (OpenAlex); provider terms (others)

The full list with versions is in the Technical & Tools Declaration.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

YIP

25/10/2026